

Fixed points, degree, index and all that...

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INTRODUCTION

Many problems in Mathematics can be formulated in terms of finding solutions to an equation of the form

$$y = f(x)$$

for some mapping f , and determining the number of its solutions. The entire field of Algebra is concerned with the specific case in which the mapping f is a polynomial of some *degree* $\deg(f) \in \mathbb{N}$, so that this becomes an algebraic problem with a number of (complex) solutions given by this integer. Notice that in principle one should usually specify a particular value of y and study how $\deg(f, y)$ varies along such values.

An important instance of this problem is the corresponding homogeneous equation:

$$f(x) = 0$$

in which we count the number $\deg(f, 0)$ of zeros of the mapping f . One interesting application of such formulation is the *fixed-point problem*:

$$x = f(x)$$

of which we would like to count the solutions via some *fixed-point index* $\text{ind}(f)$; since we can conveniently state the fixed-point problem in terms of the equivalent homogeneous equation for the perturbation of the identity $\text{Id}(x) \equiv x$ induced by f , namely: $x - f(x) = 0$, we impose:

$$\text{ind}(f) = \deg(\text{Id} - f, 0).$$

We shall now consider some examples, from a variety of fields, in which an instance of the previous problems, in particular the fixed-point formulation, plays a central role.

EXAMPLE. Consider the Initial Value Problem for trajectories $\varphi : t \in \mathbb{R} \mapsto \varphi_t \in X$, for some space X , namely $\dot{\varphi}_t = \xi(t, \varphi_t)$ for some specified initial datum φ_0 , then this Ordinary Differential Equation is equivalent to the fixed point problem

$$\varphi_t = f(\varphi_t), \quad f(\varphi_t) \equiv \varphi_0 + \int_0^t \xi(s, \varphi_s) \, ds \quad (\text{IVP})$$

having reformulated the ODE in terms of an integral equation. One usually assumes that each trajectory is a continuous function $\varphi_t \in X = C(\mathbb{R}^n)$, so we would like to have a fixed-point theory for maps on Banach spaces like the space of continuous functions.

EXAMPLE. In Game Theory, the equilibrium configurations of a system of strategic agents can be characterized as fixed points of certain mappings on the space of mixed strategies. If we let $f : \Delta \longrightarrow 2^\Delta$ denote the best-response map, associating each strategy $x \in \Delta$ in the probability simplex to the set $f(x) \subset \Delta$ of best responses to the strategy. Then, a Nash equilibrium corresponds precisely to a fixed point of this set-valued map:

$$x \in f(x)$$

meaning that the strategy x is a best response to itself. We thus need a fixed-point theory for such maps whose images of a single point are an entire set.

Here, by a *fixed-point theory* we mean the type of assumptions needed on our mapping f to determine whether fixed points exist and to determine how many there are. The two above examples respectively provide an instance of the problem where we are interested in the well-posedness, existence and uniqueness, for the fixed-point problem, and in some rule for the number of solutions.

After the development of Topology boosted by Henri Poincaré, this problem was readily identified as one of a topological nature, leading to the development of two fundamental notions in two important different branches of the subject. The existence of (more than one) fixed points and its smooth invariance is the core of the mapping degree theory in the field of Differential Topology, and it is mainly due to Brouwer. The problem of developing a theory to count fixed points, which is an algebraic matter, is just an application of the important theory of Homology in the field of Algebraic Topology: one associates to each space an algebraic structure, in

which to perform the computations. The main contributions here came from Poincaré himself, Hopf and Lefschetz.

We shall first introduce a “zoology” of classical fixed-point theorems and their applications in Chapter I, and then delve into the field of Differential and Algebraic Topology respectively in Chapter II and III.

CHAPTER I

FIXED-POINT THEOREMS

Let us first consider the easy case of a continuous function $f : [a, b] \subset \mathbb{R} \longrightarrow [a, b]$, and notice that, if $x^* \in [a, b]$ is a fixed point of f , then it is a fixed point of the continuous function $f \circ f : \mathbb{R} \longrightarrow \mathbb{R}$ too, in fact,

$$f(f(x^*)) = f(x^*) = x^*$$

and thus, iteratively, of the composition

$$\underbrace{f \circ \cdots \circ f}_{k \text{ times}}.$$

This means that we can define a sequence $\{x_k\}_{k \in \mathbb{N}}$ given by the iterative update $x_{k+1} = f(x_k)$ so that $x_k \longrightarrow x^*$ as $k \longrightarrow +\infty$. Namely, this iterative scheme will always converge to the nearest fixed point of the function.

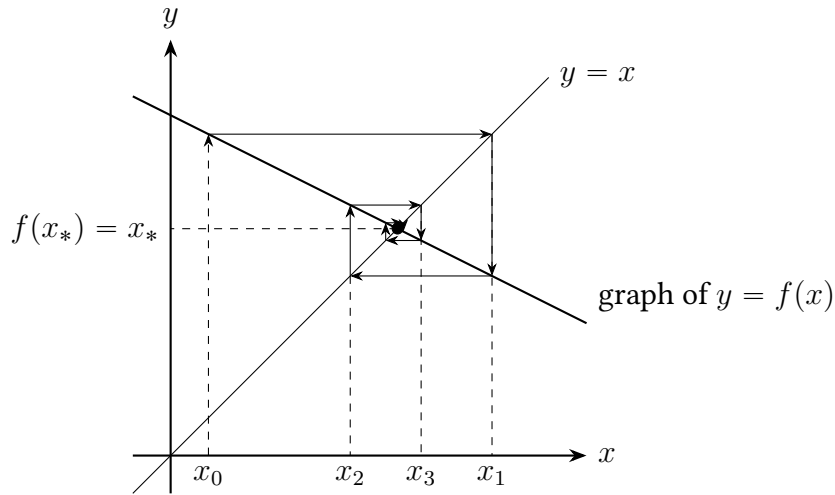


FIGURE 1. Visualization of the iterative update converging to the fixed point.

Suppose we don't know whether the function has a fixed point, then if the iterative scheme above converges to a limit, that is a fixed point. For instance, if f is increasing then $\{x_k\}_k$ is clearly monotone and it converges to a fixed point of f .

By looking at the problem geometrically, we can come up with weaker assumptions on the behaviour of the function f that ensure existence (which follows already from continuity, more precisely by the Intermediate Value Theorem) and uniqueness of a fixed point. We are looking at square $[a, b] \times [a, b] \subset \mathbb{R}^2$ and the curve $\gamma(x) = (x, f(x))$ tracing the graph of our function inside this square: fixed points correspond to the intersection of γ with the curve the bisecting line $\text{Id}(x) = (x, x)$.

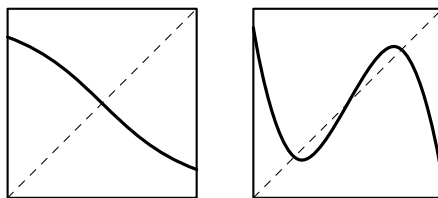


FIGURE 2. Configuration where the fixed point is unique (left) and when there are multiple intersections (right).

We notice that, if the function is “flatter” than the diagonal line, it cannot change direction fast enough to cross the line, turn around, and cross it again: it must cross exactly once. This means that if the function is a *contraction*, the fixed point is unique.

Our objective in this chapter is to first generalize the Euclidean statement of existence by continuity to arbitrary dimensions (Brouwer fixed-point theorem) and to set-valued maps (Kakutani fixed-point theorem, which will be crucial in proving existence of Nash equilibria, while in Appendix A we illustrate some nice applications to Game Theory of Brouwer's theorem itself), and then to prove the same result for compact operators in Banach spaces (Schauder fixed-point theorem) and finally to prove the celebrated contraction principle of Banach and Caccioppoli, the last two results respectively corresponding to the Peano and Picard-Lindelöf theorems for solutions to (IVP).

1. BROUWER FIXED-POINT THEOREM

There are a number of different proofs of the Brouwer fixed-point theorem using different topological and geometrical tools, some of which we shall eventually give, including Brouwer's original proof using his own concept of degree. Here however, we start by providing a fully topological proof using the concept of homology [Rot], both to highlight once more the topological nature of fixed-point theory and to have a justification for the introduction of such concepts of which we shall make a large use.

In Algebraic Topology, one can associate to any topological space X a Abelian group $H_n(X)$ such that for each continuous function, i.e. a homeomorphism $f : X \longrightarrow Y$, between topological spaces, there is a homomorphism of groups $f_* : H_n(X) \longrightarrow H_n(Y)$. The power of such a bridge (more precisely, this is called a *functor*) between a topological and algebraic structure is that one can deduce topological invariants just by looking at the properties of this groups. For instance, $H_n(\mathbb{D}^{n+1}) \cong 0$ and $H_n(\mathbb{S}^n) \cong \mathbb{Z}$ for all $n \geq 1$, i.e. the n^{th} group associated to the Euclidean unit disk $\mathbb{D}^{n+1}$ is the trivial one, but this is not the case for the unit sphere $\mathbb{S}^n = \partial\mathbb{D}^{n+1}$, for any dimension $n \geq 1$, and indeed the two spaces cannot be homeomorphic.

DEFINITION 1.1. A subspace $S \subset X$ is a *retract* of its ambient space topological space X if there exists a *retraction* $r : X \longrightarrow S$, that is a continuous map such that $r|_S \equiv \text{Id}_S$ i.e. $r(x) = x$ for all $x \in S$.

EXAMPLE. $S = \mathbb{S}^n$ is not a retract of $X = \mathbb{D}^{n+1}$ because this would require $r \circ \iota \equiv \text{Id}_S$ where $\iota : \mathbb{S}^n \hookrightarrow \mathbb{D}^{n+1}$ is the inclusion map, and thus $r_* \circ \iota_* = \text{Id}_{H_n(S)}$ i.e. the following diagram commutes:

$$\begin{array}{ccc} & & H_n(X) \cong 0 \\ & \nearrow \iota_* & \downarrow r_* \\ H_n(S) & \xrightarrow{\text{Id}} & H_n(S) \cong \mathbb{Z} \end{array}$$

but this is not the case since this would require $\text{Id} = r_* \circ \iota_* \equiv 0$.

This example actually serves as a lemma for our objective.

BROUWER FIXED-POINT THEOREM. If $f : \mathbb{D}^n \longrightarrow \mathbb{D}^n$ is continuous, then f has a fixed point.

PROOF. Suppose by contradiction that $f(x) \neq x$ for all $x \in \mathbb{D}^n$. We can thus always join the two distinct points $x, f(x) \in \mathbb{D}^n$ with a line and we let $r : \mathbb{D}^n \longrightarrow \partial\mathbb{D}^n = \mathbb{S}^{n-1}$ to be the intersection of the ray from $f(x)$ to x with the unit circle.

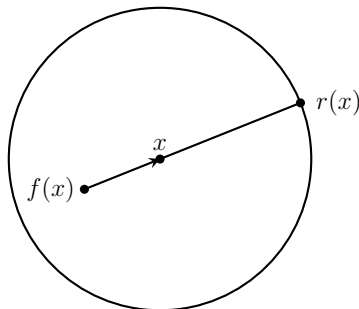


FIGURE 3. Illustration of the construction involved in the proof of the Brouwer fixed-point theorem.

This function r is then a retract, so we have contradicted the lemma. ■

The functors we introduce successfully converted a topological problem into an algebraic one, providing us with a very simplified proof. The general case for a convex, nonempty, compact set $K \subset \mathbb{R}^n$ applies because such a set is always homeomorphic to the unit disk $\mathbb{D}^n$.

COROLLARY 1.2. Let $K \subset \mathbb{R}^n$ be compact and convex and $f : K \longrightarrow K$ be continuous, then f has a fixed point.

There are a couple of important applications which follow from this theorem, which can be found in Appendix A.

2. KAKUTANI FIXED-POINT THEOREM

Before extending Brouwer to multi-valued maps, let us first provide an alternative simple proof of Brouwer's fixed point theorem using simplicial methods, restricting for simplicity to the two-dimensional case. [Zeib]

2.1 SIMPLICIAL METHODS

Let T be a closed triangle in $\mathbb{R}^2$ with vertices $p_0, p_1, p_2 \in \mathbb{R}^2$.

SPERNER LEMMA. Let the triangle T be triangulated by sub-triangles. Instead of p_0, p_1, p_2 we simply denote the vertices of T by 0, 1, 2, and assign to each knot point of the triangulation a number which belongs to some vertex of its base. Then there exists a Sperner simplex, i.e. a subtriangle having vertices with numbers 0, 1, 2.

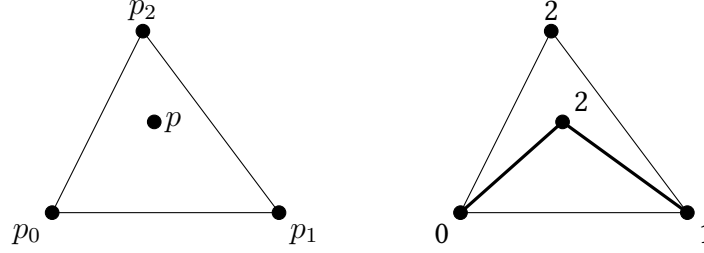


FIGURE 4. Illustration of the Sperner lemma.

KNASTER–KURATOWSKI–MAZURKIEWICZ LEMMA. Suppose a closed triangle $T \subset \mathbb{R}^2$ with vertices $p_0, p_1, p_2 \in \mathbb{R}^2$ is covered with three closed sets $C_0, C_1, C_2 \subset \mathbb{R}^2$ with

$$\text{co}(p_i, p_j, p_k) = C_i \cup C_j \cup C_k$$

for all possible index combinations, where index repetitions are allowed. Then, the intersection $C_0 \cap C_1 \cap C_2$ is not empty.

PROOF. For $k = 1, 2, \dots$ consider a sequence of triangulations of T , where the maximal diameters of the subtriangles tend to zero as $k \rightarrow +\infty$. For every triangulation there exists a Sperner simplex with vertices $p_0^{(k)}, p_1^{(k)}, p_2^{(k)}$ which carries the numbers 0, 1, 2. The construction of the labelling implies that

$$p_i^{(k)} \in C_i, \quad i = 0, 1, 2$$

by assumption. We now choose a convergent subsequence with $p_i^{(k_\ell)} \rightarrow q_i$ as $\ell \rightarrow +\infty$. Since the diameters of the subtriangles tend to zero, we obtain $q_0 = q_1 = q_2 \in C_i$. ■

By an n -dimensional closed simplex Δ we mean the convex hull of points $p_0, \dots, p_{n+1} \in \mathbb{R}^n$ which do not all lie in a hyperplane.

BROUWER FIXED-POINT THEOREM. Every continuous map $f : \Delta \rightarrow \Delta$, of an n -dimensional closed simplex Δ into itself, has a fixed point.

PROOF. Let $n = 2$, i.e. Δ is a triangle $T \subset \mathbb{R}^2$. Every point p has the representation

$$p = \lambda_0^{(p)} p_0 + \lambda_1^{(p)} p_1 + \lambda_2^{(p)} p_2$$

with barycentric coordinates:

$$0 \leq \lambda_0, \lambda_1, \lambda_2 \leq 1, \quad \lambda_0 + \lambda_1 + \lambda_2 = 1.$$

We set

$$C_i = \{p \in T : \lambda_i^{(f(p))} \leq \lambda_i^{(p)}\}.$$

so that these sets satisfy the assumptions of the KKM lemma, whence there exists a point $p \in T$ with $p \in C_0 \cap C_1 \cap C_2$. This is the required fixed point, because

$$\lambda_i^{(f(p))} \leq \lambda_i^{(p)}, \quad i = 0, 1, 2$$

is actually an equality for barycentric coordinates, and thus $f(p) = p$. ■

Corollary 1.2 also follows from the previous, as any convex compact subset $K \subset \mathbb{R}^n$ is homeomorphic to a closed simplex.

KAKUTANI FIXED-POINT THEOREM. The multivalued map $f : \Delta \longrightarrow 2^\Delta$ has a fixed point if the following conditions are satisfied:

- (1) $f(x) \subset \Delta$ is convex and nonempty for every $x \in \Delta$;
- (2) f has closed graph $\mathcal{G}(f) = \{(x, f(x)); x \in \Delta\} \subset \Delta \times \Delta$.

PROOF. Once more let $n = 2$, and T a triangle in $\mathbb{R}^2$. For $k = 1, 2, \dots$ we choose a sequence of triangulations of T , for which the maximal diameter tends to zero as $k \longrightarrow +\infty$. For each triangulation we construct maps $f_k : \Delta \longrightarrow \Delta$ by assigning to each knot point x a point $f_k(x)$ in $f(x)$, and by then extending f_k linearly to the subtriangles. According to Brouwer fixed-point theorem, each f_k has a fixed point x_k . Suppose it lies in the triangle with vertices $p_0^{(k)}, p_1^{(k)}, p_2^{(k)}$. From the definition of f_k it follows that

$$(p_i^{(k)}, f_k(p_i^{(k)})) \in \mathcal{G}(f), \quad i = 0, 1, 2$$

and there exist convergent subsequences, not separately denoted, with

$$p_i^{(k)} \longrightarrow q_i, \quad f_k(p_i^{(k)}) \longrightarrow s_i \quad \text{as } k \longrightarrow +\infty.$$

Since the triangle diameters tend to zero as $k \longrightarrow +\infty$, we obtain $q_0 = q_1 = q_2$, that is, $x_k \longrightarrow q_0$ as $k \longrightarrow +\infty$. Moreover, we have

$$x_k \in \text{co}\{p_0^{(k)}, p_1^{(k)}, p_2^{(k)}\}.$$

By using the representation of the convex hull through convex linear combinations,:

$$x_k \in \text{co}\{f_k(p_0^{(k)}), f_k(p_1^{(k)}), f_k(p_2^{(k)})\}$$

and choosing convergent subsequences of the barycentric coordinates: $q_0 \in \text{co}\{s_0, s_1, s_2\}$. Furthermore, by closedness of the graph: $(q_0, s_i) \in \mathcal{G}(f)$, i.e. $s_i \in f(q_0)$ for all i . Because of the convexity of $f(q_0)$ we have $q_0 \in f(q_0)$, i.e. q_0 is the required fixed point. ■

2.2 EXISTENCE OF NASH EQUILIBRIA

Let Δ be the space of mixed strategy profiles for N players, represented by the product of simplexes $\Delta = \prod_{i=1}^N \Delta_i$, which is a compact and convex set. The map $f : \Delta \longrightarrow 2^\Delta$ is the best-response correspondence. For a strategy profile $x \in \Delta$, the set $f(x)$ is defined as the set of strategies $y \in \Delta$ that maximize the expected utility for each player given the strategies of others. If $u_i(y_i, x_{-i})$ represents the expected utility of player i using strategy y_i against opponents x_{-i} , then:

$$f(x) = \prod_{i=1}^N \text{argmax}_{y_i \in \Delta_i} u_i(y_i, x_{-i}).$$

For any player i , the strategy space Δ_i is a simplex, which is a compact set. The expected utility function $u_i(y_i, x_{-i})$ is continuous with respect to y_i , in fact it is linear in y_i . By Weierstraß extreme value theorem, a continuous function on a compact set must attain a maximum, so the set of maximizing strategies $\text{argmax}_{y_i \in \Delta_i} u_i(y_i, x_{-i})$ is never empty. Since this holds for all players, the product set $f(x)$ is nonempty.

The expected utility $u_i(y_i, x_{-i})$ is linear in the player's own mixed strategy y_i . Let y_i^a and y_i^b be two best responses for player i to x_{-i} . This means they both yield the same maximum utility

value, denoted v^* :

$$u_i(y_i^a, x_{-i}) = v^* \quad \text{and} \quad u_i(y_i^b, x_{-i}) = v^*.$$

For any convex combination $y_i^\lambda = \lambda y_i^a + (1 - \lambda)y_i^b$ where $\lambda \in [0, 1]$, linearity implies:

$$u_i(y_i^\lambda, x_{-i}) = \lambda u_i(y_i^a, x_{-i}) + (1 - \lambda)u_i(y_i^b, x_{-i}) = \lambda v^* + (1 - \lambda)v^* = v^*.$$

Thus, y_i^λ is also a best response. This shows that for each player, the set of best responses is convex. Since the Cartesian product of convex sets is convex, $f(x)$ is a convex set.

Since $y_i^{(k)} \in f(x^{(k)})$, for every player i and for any arbitrary strategy $z_i \in \Delta_i$, the utility of playing $y_i^{(k)}$ was at least as good as playing z_i :

$$u_i(y_i^{(k)}, x_{-i}^{(k)}) \geq u_i(z_i, x_{-i}^{(k)}).$$

The utility function u_i is continuous in all arguments (both one's own strategy and opponents' strategies). Taking the limit as $k \rightarrow +\infty$ on both sides of the inequality preserves the inequality:

$$u_i(y_i, x_{-i}) = \lim_{k \uparrow +\infty} u_i(y_i^{(k)}, x_{-i}^{(k)}) \geq \lim_{k \uparrow +\infty} u_i(z_i, x_{-i}^{(k)}) = u_i(z_i, x_{-i})$$

since this inequality holds for any arbitrary alternative strategy z_i , y_i must be a best response to x_{-i} . Therefore, $y \in f(x)$, proving the graph is closed.

Since $f(x)$ is nonempty and convex, and f has a closed graph, the Kakutani fixed-point theorem applies. Thus, there exists a fixed point $x^* \in f(x^*)$, which corresponds to a Nash Equilibrium.

3. FIXED-POINT THEOREMS IN BANACH SPACES

For completeness, we now want to extend this fixed-point theory to the infinite-dimensional case, so that it holds for spaces of functions that are relevant to the theory of Dynamical Systems. [Zeia]

In the following assume X to be a Banach space or, equivalently, a complete metric space with distance

$$d(x, y) = \|x - y\|, \quad x, y \in X.$$

3.1 SCHAUDER FIXED-POINT THEOREM

In Brouwer we assume continuity, so we must find the equipollent notion for the operators $f : K \subset X \longrightarrow X$ on a Banach space, where K is taken to be a nonempty convex subset. Assuming moreover closedness of the domain K , for finite-dimensional Banach spaces, continuous and compact operators are the same. It turns out that is the right assumption.

SCHAUDER FIXED-POINT THEOREM. If $K \subset X$ is a convex closed subset of a Banach space X and $f : K \longrightarrow K$ is a compact operator, then f has a fixed point.

The following, can be seen as a corollary of the Schauder fixed-point theorem.

PEANO THEOREM. Let $Q \subset \mathbb{R} \times \mathbb{R}^n$ be a closed rectangle with $(t_0, \varphi_0) \in \text{int } Q$, and let $\xi : Q \longrightarrow \mathbb{R}^n$ be a function that is continuous on Q . Then there exists some $\epsilon > 0$ such that (IVP) has at least one solution on the interval $[t_0 - \epsilon, t_0 + \epsilon]$.

Assuming continuity, one gets local existence of solutions to the ODE: $\dot{\varphi}_t = \xi(t, \varphi_t)$.

3.2 THE CONTRACTION PRINCIPLE

For completeness, we here sketch the Banach theory of existence and uniqueness of fixed points when the mapping is a contraction. That corresponds to the additional assumption of Lipschitz continuity in ODEs, leading to local existence and uniqueness for such problem.

BANACH-CACCIOPPOLI FIXED-POINT THEOREM. Let (X, d) be a complete metric space with a *contraction mapping* $f : X \longrightarrow X$, i.e.

$$d(f(x), f(y)) \leq \lambda d(x, y)$$

for all $x, y \in X$ and some fixed $\lambda \in [0, 1)$. Then f admits a unique fixed-point $x^* \in X$.

Contractivity allows the fixed point iteration $x_{k+1} = f(x_k)$ to always converge, exactly to the fixed point.

The following, can be seen as a corollary of the Banach-Caccioppoli fixed-point theorem.

PICARD-LINDELÖF THEOREM. Let $Q \subset \mathbb{R} \times \mathbb{R}^n$ be a closed rectangle with $(t_0, y_0) \in \text{int } Q$, and $\xi : Q \longrightarrow \mathbb{R}^n$ be a function that is continuous in t and Lipschitz continuous in φ (with Lipschitz constant independent from t). Then there exists some $\epsilon > 0$ such that (IVP) has a unique solution on the interval $[t_0 - \epsilon, t_0 + \epsilon]$.

3.3 FIXED POINTS OF MULTIVALUED MAPS

Given a set-valued map $f : X \longrightarrow 2^X$, one can easily generalize the concept of contractivity for such operators by means of the distance between subsets $A, B \subset X$ defined as follows:

$$d(A, B) = \max \left\{ \sup_{a \in A} d(a, B), \sup_{b \in B} d(A, b) \right\}$$

where $d(x, K) = \inf_{y \in K} d(x, y)$ for any $K \subset X$. Thus, the Banach fixed-point theorem also generalizes to this setting in a straightforward way.

The Brouwer-Kakutani-Schauder statement of existence here instead requires the additional assumption that the preimages $f^{-1}(y) \subset K$ are open for any $y = f(x) \in K$. An interesting application is the existence of saddle points for a specific kind of functions: that is, the infinite-dimensional generalization of von Neumann's Minimax theorem.

MINIMAX THEOREM. Let $A, B \subset X$ be convex, closed and bounded, and let $f : A \times B \longrightarrow \mathbb{R}$ be a function that is concave on A and convex on B , then there exists a saddle point, at which

$$\min_A \max_B f = \max_B \min_A f.$$

SKETCH OF THE PROOF. Because $\min_A f \leq \min_A \max_B f$, we need only to show that

$$\min_A \max_B f < \max_B \min_A f.$$

Define the operator $F : A \times B \longrightarrow 2^{A \times B}$ given by

$$F(a, b) = \{(u, v) \in A \times B : f(u, b) < t, f(a, v) > s\}, \quad a \in A, b \in B$$

where $t + \epsilon = \max_B \min_A f$ and $s + \epsilon = \min_A \max_B f$ for some fixed $\epsilon > 0$. Each image $F(a, b) \subset A \times B$ is convex and preimage $F^{-1}(u, v)$ open. We can conclude there exists a fixed point $(a, b) \in F(a, b)$ for which

$$\min_A \max_B f(a, b) - \epsilon = s < f(a, b) < t = \max_B \min_A f + \epsilon$$

and we get our claim by letting the arbitrary $\epsilon \downarrow 0$. ■

The equilibrium (saddle point) in a minimax problem corresponds to a fixed point of a best-response mapping, with the minimax equality ensuring that no player can unilaterally deviate to improve their payoff, which is exactly the Nash condition.

More formally, if $f_i(y_i, x_{-i}) = -u_i(y_i, x_{-i})$ is the player's loss function, then the Nash equilibrium corresponds to:

$$\min_{y_i \in \Delta_i} \max_{x_{-i} \in \Delta_{-i}} f_i(y_i, x_{-i}) = \max_{x_{-i} \in \Delta_{-i}} \min_{y_i \in \Delta_i} f_i(y_i, x_{-i}),$$

so that no deviation can yield a strictly better outcome.

CHAPTER II

MAPPING DEGREE THEORY

Let us now study the number of solutions to the equation $y = f(x)$ where $f : [a, b] \subset \mathbb{R} \longrightarrow \mathbb{R}$ is a differentiable function with $f(a), f(b) \neq 0$, and start by once more looking at the fixed point problem: since now f takes arbitrary values in $\mathbb{R}$, continuity is not any more a sufficient for having fixed point: suppose $f(a) > a$, then we want contractivity (if, for instance, we have $|f'| = 1$ then the graph of the function will be a line parallel to $y = x$ and they will never intersect) while if $f(a) < a$, the function needs to have a superlinear growth in some part of the interval for being able to reach and touch $y = x$. In that case, we would have a fixed point: to have one more, the function must change direction by reaching a maximum (or minimum, in the first instance) and changing the sign of its derivative. We have thus reduced the problem to counting the zeros of the derivative of f . The crucial fact here is that we don't care at which value the function has a fixed point.

EXAMPLE. The function $f(x) = x^2 - ax$ with $a > 0$ always has three fixed point, however so does the shifted function $\tilde{f}(x) = f(x) + b = x^2 - ax + b$ with $0 < b \leq a$. So, we would like the degree to be invariant under such transformations. [Zeia]

The most simple example of this invariance is that of the zeros of $f(x) = x$, as there always is one no matter the affine transformation we apply to the identity function: $\deg(\text{Id}, 0) = 1$. The same holds for $f(x) = x^3$, however for that the situation is more delicate as f has a degenerate zero at $x = 0$ (i.e. $f'(0) = 0$) which disappears under these affine transformations. In general, we can modify any function so that the new function has no zeros at the boundary points and, in the interior (a, b) of the interval it has no or finitely many regular zeros $\{z_i\}_i$, in which case we can set $\deg(f, 0) = \sum_i \text{sign}(f'(z_i))$.

We may also want to consider more interesting problems, for instance:

$$y = f(x), \quad f : [a, b] \subset \mathbb{R} \longrightarrow \mathbb{R}^2$$

so that we are looking at a curve $\gamma(t) = (t, f(t))$ parametrized by $t \in \mathbb{R}$. Suppose $f(a) = f(b)$ so that γ is closed and we can identify our domain $[a, b]$ with the circle $\mathbb{S}^1$: assuming that the entire image γ is included in the circle, we are basically considering an endomorphism $\vartheta \mapsto f(\cos \vartheta, \sin \vartheta) = f(e^{i\vartheta})$ of the manifold $\mathbb{S}^1$.

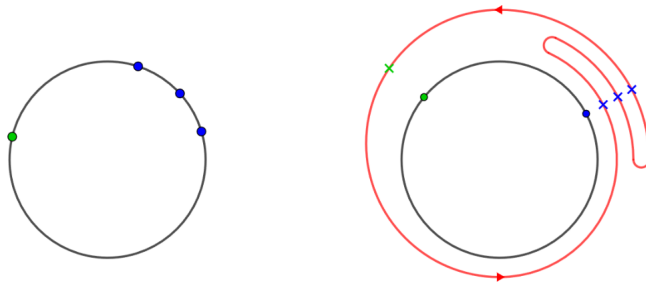


FIGURE 1. Illustration of a non-injective map $f : \mathbb{S}^1 \longrightarrow \mathbb{S}^1$, for instance the three regular points on the left are all mapped to the blue value on the right. Let y be the green regular value on the right, then clearly $\deg(f, y) = 1$. Similarly for the blue regular value, which is the image of three different points but is reached twice with a positive (counterclockwise) orientation and once clockwise: summing $1 - 1 + 1 = 1$ is still the degree of f at this value.

In this case, critical points/values still correspond to a zero in the derivative, i.e. the velocity $\dot{\gamma}(t)$ of the curve we are looking. However, here it doesn't make sense to speak of the sign of the derivative as, at each zero of the velocity, it is the orientation of curve that changes.

The lesson to learn here is threefold: the degree of a smooth function does not depend on the value y at which we compute it, we need to look only at the regular values of the function and the right place to do this is on a smooth manifold.

1. REGULAR VALUES AND SARD THEOREM

Let $\mathcal{M}$ be a smooth manifold of dimension m and consider smooth curves $\gamma : \mathbb{R} \longrightarrow \mathcal{M}$ assigning to each time a point $\gamma(t) \in \mathcal{M}$ and tangent vector $\dot{\gamma}(t)$ to the manifold. Take any point $p \in \mathcal{M}$ and curves for which $\gamma(0) = p$; if we identify all such curves by the following equivalence

$$\gamma_1 \sim \gamma_2 \quad \text{if} \quad \dot{\gamma}_1(0) = \dot{\gamma}_2(0)$$

then the set of all the equivalence classes is called the tangent space $T_p\mathcal{M}$ of the manifold at the point p and consists of all initial velocities of curves originating at that point. Given any smooth function $f : \mathcal{M} \longrightarrow \mathcal{N}$ between manifolds, it then induces a linear map

$$df_p : T_p\mathcal{M} \longrightarrow T_{f(p)}\mathcal{N}$$

given by

$$df_p(v) = \left. \frac{d}{dt}(f \circ \gamma) \right|_{t=0} \quad \text{if } v = \dot{\gamma}(0) \in T_p\mathcal{M}$$

which effectively gives the differential of the smooth map considered, as the composition $f \circ \gamma$ creates a curve in $\mathcal{N}$ passing through $f(p)$. [Lee]

DEFINITION 1.1. We say that $p \in \mathcal{M}$ is a *regular point* of f if df_p is invertible. If the differential of f at p is instead singular, we say that it is a *critical point* of f . We denote by $\mathcal{R}_f, \mathcal{C}_f \subset \mathcal{M}$ respectively the set of regular and critical points of f .

If $p \in \mathcal{M}$ is a regular point, then it follows by the Inverse Function Theorem that f maps some neighbourhood of p diffeomorphically onto some open set in $\mathcal{N}$.

DEFINITION 1.2. The point $q \in \mathcal{N}$ is a *regular value* if $f^{-1}(q) \subset \mathcal{M}$ only contains regular points, i.e. $f^{-1}(q) \subset \mathcal{R}_f$.

EXAMPLE. When $\mathcal{M} = \mathcal{N} = \mathbb{R}$, for any smooth function its set $\mathcal{R}_f$ of critical values always consists of a discrete collection of countably many points, which is a negligible set in $\mathbb{R}$. The same can be said for the case $\mathcal{M} = \mathbb{R}$ and $\mathcal{N} = \mathbb{R}^n$ as can be observed in Figure 1 (which corresponds to $n = 2$), and this is equivalent to saying that there do not exist space-filling curves, differently from the non-smooth case.

SARD THEOREM. Given a smooth map $f : \mathcal{M} \longrightarrow \mathcal{N}$, then the set $f(\mathcal{R}_f) \subset \mathcal{N}$ of its regular values is dense.

Please refer to Appendix B for the proof of this deep result, which fully justifies our restricting to only its regular values when studying the degree of a map.

2. THE DEGREE MODULO 2 OF A MAPPING

Consider once more $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$. This has a critical point at $x = 0$ so we perturb it and consider instead $\tilde{f}(x) = x^2 - \epsilon$ for some $\epsilon \neq 0$. For $\epsilon > 0$, the map $\tilde{f}$ has two zeros $z = \pm\sqrt{\epsilon}$ so that $\deg(f, 0) = \text{sign}(\tilde{f}'(\sqrt{\epsilon})) - \text{sign}(\tilde{f}'(-\sqrt{\epsilon})) = 1 - 1 = 0$. But the result is the same for $\epsilon < 0$, since in that case the perturbed function has no zeros. If y is a regular value, recall that $\#f^{-1}(y)$ denotes the number of solutions x to the equation $y = f(x)$ and we claim that:

$$\#f^{-1}(y) \equiv \#\tilde{f}^{-1}(y) \pmod{2}$$

for the type of transformations $\tilde{f}$ we have thus far considered and that we shall now make precise. [Mil]

DEFINITION 2.1. Two mappings $f, \tilde{f} : \mathcal{M} \rightarrow \mathcal{N}$ are said to be *smoothly homotopic*: $f \sim \tilde{f}$, if there exists a smooth homotopy $H : \mathcal{M} \times [0, 1] \rightarrow \mathcal{N}$ between them, i.e. a map such that:

$$H(p, 0) = f(p), \quad H(p, 1) = \tilde{f}(p)$$

for all $p \in \mathcal{M}$.

HOMOTOPY LEMMA. Let $f, \tilde{f} : \mathcal{M} \rightarrow \mathcal{N}$ be smoothly homotopic maps between manifolds of the same dimension and $q \in \mathcal{N}$ to be a regular value of both f and g , then

$$\#f^{-1}(q) \equiv \#\tilde{f}^{-1}(q) \pmod{2}.$$

PROOF. Let $H : \mathcal{M} \times [0, 1] \rightarrow \mathcal{N}$ be the smooth homotopy between f and $\tilde{f}$. We distinguish cases: if q is a regular value for $H_t, t \in [0, 1]$, then $H_t^{-1}(q) \subset \mathcal{R}_{H_t}$ is a compact and connected one-dimensional manifold with boundary given by the set

$$H_t^{-1}(q) \cap (\mathcal{M} \times \{0\} \cup \mathcal{M} \times \{1\}) = (f^{-1}(q) \times \{0\}) \cup (\tilde{f}^{-1}(q) \times \{1\})$$

which contains exactly $\#f^{-1}(q) + \#\tilde{f}^{-1}(q)$ points. But a compact and connected one-dimensional manifold with boundary is diffeomorphic to $[0, 1]$, whence it has an even number of boundary points, i.e. $\#f^{-1}(q) + \#\tilde{f}^{-1}(q)$ is even, which is our claim.

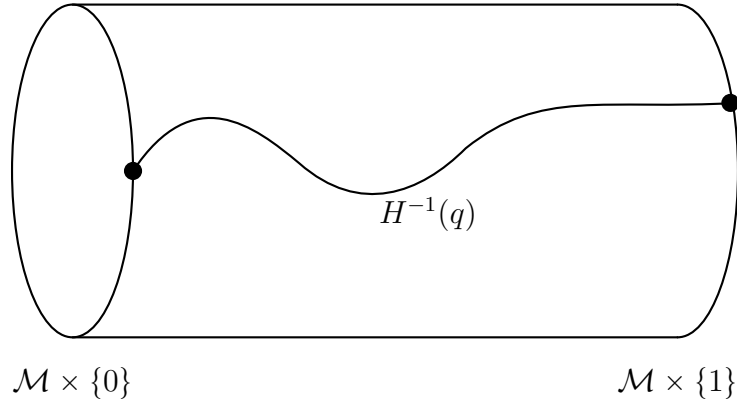


FIGURE 2. The number of boundary points on the left is congruent to the number on the right modulo 2. [Mil]

If instead q is not a regular value for $H_t, t \in [0, 1]$, we can find distinct neighbourhoods in $\mathcal{N}$ over which $\#f^{-1}(q), \#\tilde{f}^{-1}(q)$ are locally constant, so it is enough to find a regular value of the homotopy within the intersection of these two neighbourhoods to complete the proof. ■

THEOREM 2.2. If p, q are regular values of $f : \mathcal{M} \longrightarrow \mathcal{N}$, then:

$$\#f^{-1}(p) \equiv \#f^{-1}(q) \pmod{2}.$$

PROOF. Let $h : \mathcal{N} \longrightarrow \mathcal{N}$ be a diffeomorphism which is isotopic to the identity and which carries p onto q . Then q is a regular value of the composition $h \circ f : \mathcal{M} \longrightarrow \mathcal{N}$. Since $h \circ f$ is homotopic to f , by the Homotopy lemma:

$$\begin{aligned} \#f^{-1}(q) &\equiv \#(h \circ f)^{-1}(q) \pmod{2} \\ &\equiv \#f^{-1}(h^{-1}(q)) \pmod{2} \\ &\equiv \#f^{-1}(p) \pmod{2}, \end{aligned}$$

as required. ■

We call this common residue class $\deg_2(f)$ the *degree modulo 2 of the mapping*, which does not depend on the specific values taken by f but only on its smooth homotopy class. In fact,

by Sard theorem there exists a regular value $q \in \mathcal{N}$ of both $f \sim \tilde{f}$ and the congruence

$$\deg_2(f) \equiv \#f^{-1}(q) \equiv \#\tilde{f}^{-1}(q) \equiv \deg_2(\tilde{f}) \pmod{2}$$

shows that $\deg_2$ is a smooth homotopy invariant.

EXAMPLE. Any constant map has even mod 2 degree, while the identity map has odd degree. It follows that the identity map $\text{Id} : \mathcal{M} \rightarrow \mathcal{M}$ of a compact boundaryless manifold is not homotopic to a constant. This fact yields another proof of Brouwer fixed-point theorem, being equivalent to the statement that the sphere is not a smooth retract of the disk, i.e. that no smooth map $f : \mathbb{D}^{n+1} \rightarrow \mathbb{S}^n$ leaves the sphere pointwise fixed, for such a map would give rise to a smooth homotopy $H_t(x) = f(tx)$ between a constant map and the identity.

3. THE BROUWER DEGREE

As we said, for generic manifolds we want to count the number of times the mapping considered reverts its orientation.

Recall that each n -dimensional vector space has precisely two orientations, corresponding to the number of equivalence classes of ordered bases defined as follows: $(v_1, \dots, v_n)$ determines the same orientation as the basis $(w_1, \dots, w_n)$ if $w_i = \sum_j a_{ij}v_j$ with $\det(a_{ij}) > 0$. Any invertible linear map between vector spaces yields a map between the orientations of the two space, effectively yielding a new orientation by means of the image vectors for the ordered basis considered.

An oriented smooth manifold consists of a manifold $\mathcal{M}$ together with a choice of orientation for each tangent space $T_p\mathcal{M}$, with the compatibility condition that there exists a diffeomorphism locally mapping neighbourhoods of each $p \in \mathcal{M}$ onto an open subset of $\mathbb{R}^m$ and which is orientation-preserving, in the sense that each isomorphism dh_p carries the specified orientation for $T_p\mathcal{M}$ into the standard orientation for $\mathbb{R}^m$. This definition does not imply that we can find a basis for each tangent space in a continuous way: for instance, we will show with the hedgehog theorem that this cannot be done on the sphere $\mathbb{S}^2$, even though it is an orientable manifold because it admits an orientation.

DEFINITION 3.1. Given a smooth map $f : \mathcal{M} \longrightarrow \mathcal{N}$ between oriented manifolds, we define the *sign* of $df_p : T_p\mathcal{M} \longrightarrow T_{f(p)}\mathcal{N}$ to be either +1 or -1 according as to whether df_p preserves or reverses orientation. For any regular value $q \in \mathcal{N}$, we also define its *degree* to be the integer given by:

$$\deg(f, q) = \sum_{p \in f^{-1}(q)} \text{sign}(df_p)$$

In essence, the degree counts how many times a point q is covered by the map f , but taking into account the orientation.

Based on the homotopic invariance of $\deg_2$, assuming that $\mathcal{N}$ is connected, we have

$$\deg(f, q_1) = \deg(f, q_2)$$

for any two regular values $q_1, q_2 \in \mathcal{N}$, and thus we shall refer solely to the *Brouwer degree*

$$\deg(f) \in \mathbb{Z}$$

of the smooth map. As a corollary of this fact, we have that if $\deg(f) \neq 0$, then f is surjective. The Brouwer degree itself inherits invariance under smooth homotopies of the degree mod 2.

EXAMPLE. Given an integer $k \in \mathbb{Z}$, consider the smooth map $f : \mathbb{S}^1 \longrightarrow \mathbb{S}^1$ given by

$$f(\cos \vartheta, \sin \vartheta) = (\cos(k\vartheta), \sin(k\vartheta)) = e^{ik\vartheta}.$$

If $k = 0$ then the map is constant, it is not surjective and its degree is necessarily 0. Assuming $k \neq 0$, firstly we notice that all the points $p = (\cos \vartheta_p, \sin \vartheta_p) \in \mathbb{S}^1$ are regular points for f , as each $T_p\mathbb{S}^1$ is spanned by the tangent vector $v_p = (-\sin \vartheta_p, \cos \vartheta_p)$ and

$$df_p(v_p) = df_p(-\sin \vartheta_p, \cos \vartheta_p) = \frac{d}{d\vartheta}(\cos(k\vartheta), \sin(k\vartheta)) \Big|_{\vartheta=\vartheta_p} = k(-\sin \vartheta_p, \cos \vartheta_p) \neq 0$$

and moreover $\text{sign}(df_p) = \text{sign}(k)$. Since any point $q \in \mathbb{S}^1$ has $|k|$ points in its preimage:

$$\deg(f) = \deg(f, p) = \sum_{p \in f^{-1}(q)} \text{sign}(df_p) = \sum_{p \in f^{-1}(q)} \text{sign}(k) = |k| \text{sign}(k) = k$$

where we have chosen an arbitrary point $p \in \mathbb{S}^1$ for the computation. Notice that here the degree is effectively counting the *winding number* of how many times the image curve of our

map “wraps” around the circle and thus how many times it attains each value $q \in \mathbb{S}^1$. Notice that here a negative k means that the map is reversing the orientation and thus the degree is negative. For instance, what we have shown is equivalent to the fact that the k^{th} power map $\zeta_k : \mathbb{S}^1 \subset \mathbb{C} \longrightarrow \mathbb{S}^1$ given by $\zeta_k(z) = z^k$ has $\deg(\zeta_k) = k$, then the conjugation map $\zeta_{-1}(z) = z^*$, reflecting the circle across the abscissæ, has $\deg(\zeta_{-1}) = -1$, as it effectively reverts the orientation. The *antipodal map*

$$\alpha : \mathbb{S}^1 \subset \mathbb{C} \longrightarrow \mathbb{S}^1, \quad \alpha(z) = -z$$

being equivalent to a rotation by $e^{i\pi}$, instead has $\deg(\alpha) = 1$. [Ful]

We can now deduce a couple of important examples of the type of general results provable using the degree and its homotopy invariance.

THEOREM 3.2. Let $f : \mathbb{S}^1 \longrightarrow \mathbb{S}^1$ be continuous. If $\deg(f) \neq 1$, then f has a fixed point.

PROOF. Assume by contradiction that f has no fixed points. Then, for every $z \in \mathbb{S}^1$, the line segment in $\mathbb{C}$ from $f(z)$ to $-z = \alpha(z)$ does not pass through the origin. Thus we can define a homotopy between f and the antipodal map α by

$$H_t(z) = \frac{(1-t)f(z) - tz}{|(1-t)f(z) - tz|}$$

but the antipodal map has degree $1 = \deg(f)$. ■

This result is interesting because it is the first fixed-point theorem involving a manifold with no vector space structure and which is not convex or simply connected (in other terms, this is the first fixed-point theorem, only assuming continuity, which cannot be seen as a generalization of Brouwer.)

We have seen that the algebraic notion of degree of a polynomial and its topological Brouwer degree do not correspond, however that is the case in the complex field $\mathbb{C}$.

FUNDAMENTAL THEOREM OF ALGEBRA. Every nonconstant complex polynomial $p(z)$ has at least one root in $\mathbb{C}$.

PROOF. Given a polynomial $p(z) = z^k + a_{k-1}z^{k-1} + \cdots + a_1z + a_0$ of degree $k \geq 1$ with complex coefficients, we assume by contradiction that it never vanishes. For each $R \geq 0$, we

can then consider the map

$$f_R : \mathbb{S}^1 \subset \mathbb{C} \longrightarrow \mathbb{S}^1, \quad f_R(e^{i\vartheta}) = \frac{p(Re^{i\vartheta})}{|p(Re^{i\vartheta})|}$$

so that, by smooth homotopic invariance, $\deg(f_R)$ does not depend on R . For $R = 0$, the map is constant and hence it is not surjective, giving $\deg(f_R) = \deg(f_0) = 0$ for all R . On the other hand, for R large enough, the map f_R is smoothly homotopic to ζ_k so it has a degree $k \neq 0$, giving a contradiction. ■

Finally, we claimed that we cannot find a basis for each tangent space $T_p\mathbb{S}^2$ in a continuous way, highlighting how this is a different notion from that of orientability. We will prove a more general statement: if n is even, then there cannot exist a smooth vector field $v : \mathbb{S}^n \longrightarrow \mathbb{R}^{n+1}$ associating to each point $p \in \mathbb{S}^n$ a nonzero tangent vector $v_p \in T_p\mathbb{S}^n$. There will always be a point at which the vector field vanishes and thus, more pictorially, the “hairy” sphere $\mathbb{S}^n$ cannot be “combed” without creating a cowlick.

HEDGEHOG THEOREM. The sphere $\mathbb{S}^n$ admits a smooth field of nonzero tangent vectors if and only if n is odd.

PROOF. The tangent space $T_p\mathbb{S}^n = \{v \in \mathbb{R}^{n+1} : p \cdot v = 0\}$ whence, given any smooth vector field $v_p \in T_p\mathbb{S}^n$, $p \cdot v_p = 0$ when seen as vectors in $\mathbb{R}^{n+1}$. Suppose without loss of generality $v_p \cdot v_p = 1$ as we can always consider the normalization $v_p/|v_p|$, and consider the map

$$H : \mathbb{S}^n \times [0, \pi] \longrightarrow \mathbb{S}^n, \quad H_\vartheta(p) = p \cos \vartheta + v_p \sin \vartheta$$

so that $H_\vartheta(p) \cdot H_\vartheta(p) = 1$ and

$$H_0(p) = p, \quad H_\pi(p) = -p = \alpha(p),$$

thus establishing smooth homotopy between the antipodal map and the identity of $\mathbb{S}^n$: this is impossible when n is even. ■

We shall see that the hedgehog theorem and theorem 3.2 are respectively an instance of the more powerful Poincaré-Hopf and Lefschetz fixed-point theorems.

3.1 MEAN-VALUE FORMULA

Given two oriented manifolds $\mathcal{M}$ and $\mathcal{N}$ of the same dimension and a smooth $F : \mathcal{M} \longrightarrow \mathcal{N}$ between them, we actually have that $\text{sign}(dF_p) = \text{sign}(\det dF_p)$, where dF_p is identified with the matrix representing the differential in terms of the chosen basis for our couple of tangent spaces. Assuming that $\mathcal{M}$ is compact and $\mathcal{N}$ is connected, we then have the following change of coordinate formula:

$$\int_{\mathcal{N}} \sum_{p \in F^{-1}(q)} f(p) = \int_{\mathcal{M}} f(p) dF_p$$

for any bounded map $f : \mathcal{M} \longrightarrow \mathbb{R}$. If we let $f(p) = \text{sign}(dF_p)$, then we get:

$$\int_{\mathcal{N}} \sum_{p \in F^{-1}(q)} \text{sign}(dF_p) = \int_{\mathcal{M}} \text{sign}(dF_p) dF_p = \int_{\mathcal{M}} \text{sign}(\det dF_p) |\det dF_p| = \int_{\mathcal{M}} \det dF_p,$$

as well as:

$$\int_{\mathcal{N}} \sum_{p \in F^{-1}(q)} \text{sign}(dF_p) = \int_{\mathcal{N}} \deg(F) = \deg(F) \text{vol}(\mathcal{N})$$

so we conclude

$$\deg(F) = \frac{1}{\text{vol}(\mathcal{N})} \int_{\mathcal{M}} \det dF_p$$

leading to the interpretation of the degree of a map as the mean multiplicity with which it covers its target manifold, being it given by ratio of the total signed volume of the image and the entire volume of the image ambient space.

The one dimensional version of this formula is the mean-value theorem for a smooth map $F : [a, b] \subset \mathbb{R} \longrightarrow [c, d] \subset \mathbb{R}$. In fact, when F is onto, then

$$\int_a^b \det dF_p = F(b) - F(a) = d - c$$

where we argue by the fundamental theorem of calculus because here $\det dF_p = F'(p)$ for each $p \in [a, b]$. But if, accounting for the sign, F covers a number $\deg(F)$ of times the image interval, then indeed:

$$\deg(F) = \frac{1}{d - c} \int_a^b \det dF_p.$$

CHAPTER III

FIXED-POINT INDEX

Given a smooth map $f : \mathcal{M} \longrightarrow \mathcal{M}$, we know want to study the number of its fixed points, i.e. its index $\text{ind}(f) = \deg(\text{Id} - f)$. Assume that $\mathcal{M} \subset \mathbb{R}^m$ is a compact with boundary and f has no fixed points in $\partial\mathcal{M}$, then we want to look at the zeros of the field $v_p = p - f(p)$ composed of the vectors joining each point $p \in \mathcal{M}$ to its image $f(p) \in \mathcal{M}$. That is, without loss of generality as in the proof of the hedgehog theorem, we consider the vector field $v_p/|v_p|$ so that on the boundary it corresponds to the the *Gauss map* $G : \partial\mathcal{M} \longrightarrow \mathbb{S}^{m-1}$, associating to each point $p \in \partial\mathcal{M}$ the outward unit normal vector $n_p \in T_p\mathcal{M}$. Through this map, one can associate to each point the *Gauss curvature* $K_p = \det dG_p$. As a consequence of the mean-value formula, we call the degree of this map

$$\deg(G) = \frac{1}{\text{vol}(\mathbb{S}^{m-1})} \int_{\mathcal{M}} \det dG_p = \frac{\Gamma(m/2)}{2\pi^{m/2}} \int_{\mathcal{M}} K_p$$

the *curvatura integra* of the manifold, as it is given by the average of the pointwise Gauss curvature. [Mil]

1. INDICES OF VECTOR FIELDS

DEFINITION 1.1. Given a smooth vector field $v : \mathcal{M} \longrightarrow \mathbb{R}^m$, with an isolated zero at the point $z \in \mathcal{M}$ and consider for any $p \in \mathcal{M}$ close to such zero the mapping $p \longmapsto v_p/|v_p| \in \mathbb{S}^{m-1}$. Then, the *index* $\text{ind}_z(v)$ of the vector field at the zero is the degree of this mapping.

EXAMPLE. For vector fields in the plane, by keeping in mind the winding number interpretation of the degree, one can easily construct examples of index 1 at 0: it is enough to consider the vortex

$$V(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

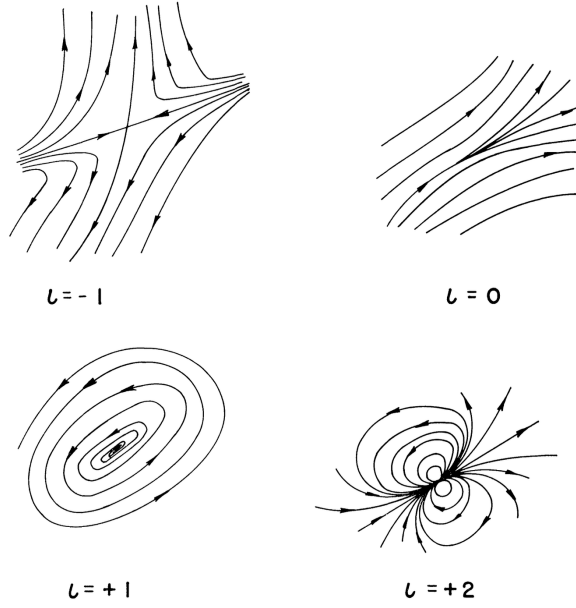


FIGURE 1. Examples of plane vector fields and their index ι at the singularity. [Mil]

or any field with a center, and similarly of index 2 by taking any vector field that traces a figure 8 around, such as $(x^2 - y^2, 2xy)$ rotating thus twice around the origin. Any vector field with a saddle point has index -1.

Interesting applications include the gradient field of some smooth function $f : \mathcal{M} \rightarrow \mathbb{R}$ having, at each nondegenerate critical point of f , either an index of 1, if f attains a local maximum or minimum, or -1 otherwise, i.e. at a saddle point. [Ful]

EXAMPLE. For vector fields in the sphere, the situation is also simple: as we have seen, singularities exist in abundance, especially for even dimension. Let us, for instance, consider the rotation field $R(x, y, z) = (-y, x, 0)$ when restricted to the sphere $\mathbb{S}^2$. Along each latitude circle, the vector points eastward and has constant length, the flow lines are parallels of latitude and, at the poles, the field vanishes, each pole behaving as a simple (elliptic) singularity with index +1, thus $\sum_z \text{ind}_z(R) = 2$.

The surprising fact is that this is the case for any vector field on $\mathbb{S}^2$.

LEMMA 1.2. For any vector field V with isolated singularities z on $\mathbb{S}^2$, $\sum_z \text{ind}_z(V) = 2$.

PROOF. Given our previous example, it suffices to show that the sum of the indices of any two such vector fields V_1 and V_2 is the same. Define the smooth homotopy of vector fields:

$$V_t = (1 - t)V_1 + tV_2, \quad t \in [0, 1]$$

which is itself a collection of vector fields whose zeros can only be created or annihilated in pairs of opposite index, by smooth homotopy invariance of $\deg_2$. Thus, the total sum of indices over all zeros of V_t remains constant as t varies continuously. Hence,

$$\sum_z \text{ind}(V_1) = \sum_z \text{ind}(V_t) = \sum_z \text{ind}(V_2), \quad t \in [0, 1]$$

by invariance of the degree under smooth homotopies. ■

Let us now look at some more strange manifold, for instance the one given by the surface of a doughnut, i.e. the two-dimensional torus $\mathbb{T}^2$. This can be realized concretely in several ways: as a surface of revolution, by an explicit equation in $\mathbb{R}^3$, as a Cartesian product $\mathbb{S}^1 \times \mathbb{S}^1$, or by taking a square and identifying opposite edges.

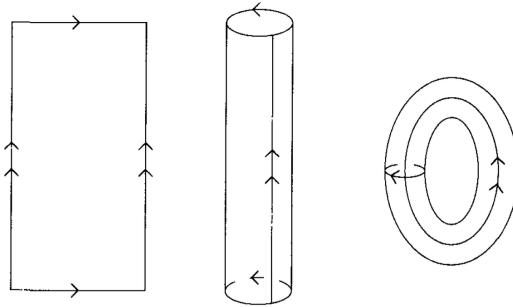


FIGURE 2. As we get the ring by identifying the endpoints of any interval $[a, b] \subset \mathbb{R}$, i.e. $\mathbb{S}^1 = \mathbb{R}/\mathbb{Z}$, similarly we get the torus by identifying opposite edges of a square in $\mathbb{R}^2$, i.e. $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{R}^2/\mathbb{Z}^2$ as we may think of it as the space of points on a square where moving one unit horizontally or vertically brings us back to where we started. [Ful]

LEMMA 1.3. For any vector field V with isolated singularities z on $\mathbb{T}^2$, $\sum_z \text{ind}_z(V) = 0$.

It is now clear that the sum of the index sum of a vector field does not really depend on the field considered but rather only on the topology of the manifold.

LEMMA 1.4. For any vector field v on $\mathcal{M}$ with only nondegenerate zeros, its index sum is equal to the curvatura integra of the manifold, i.e. $\sum_z \text{ind}_z(v) = \deg(G)$.

PROOF. Around each zero $z \in \mathcal{M}$, choose a small neighbourhood $\mathcal{O}_z \subset \mathcal{M}$ such that z is the only zero in it. Consider the manifold with holes $\mathcal{M}' = \mathcal{M} \setminus \bigcup_z \mathcal{O}_z$, on which boundary $\partial\mathcal{M}' = \bigcup_z \partial\mathcal{O}_z$, the Gauss map G is defined and extends to the interior. Since $\mathcal{M}'$ contains no zeros of v , G is homotopic to a constant in the interior and thus

$$\deg(G) = \sum_z \deg(G|_{\mathcal{O}_z}) = \sum_z \text{ind}_z(v).$$

by additivity of the degree. ■

Relating this result involving the curvature is rather cumbersome, especially as we want to consider arbitrary manifolds. Going back to our examples involving the sphere and torus, we can try to find a pattern in order to identify another topological invariant that can serve our purpose. When dealing with surfaces, if we indicate by g the number of holes, i.e. the *genus* of the surface considered, we notice that the index sum is exactly equal to $2 - 2g$: in fact, $\mathbb{T}^2$ has one hole and a genus of 0, while the sphere has no holes and thus has a genus of 2. For instance, a pretzel has a genus of 3.

Of course, we need to define precisely what an hole is (also in higher dimensions) and we would like this quantity to be in some way related to the curvatura integra.

1.1 POINCARÉ-HOPF THEOREM

For a compact surface, the idea is to construct a *triangulation*, i.e. to cut the surface into pieces homeomorphic to triangles, fitting together along the edges. By just looking at the union of the triangles, we will then have a polyhedron with a certain number V of vertices (points), a number E of edges (homeomorphic to closed intervals), and a number F of faces (homeomorphic to closed triangles). In this way, we can associate to each surface its *Euler characteristic*

$$\chi = V - E + F.$$

When we identified the surface of the torus with a square convolved twice on itself by means of the identification of opposing edges, we were implicitly looking at a couple of triangles (united in a square, as in Fig. 3) and indeed $V = 1$, $E = 3$ and $F = 2$, whence $\chi = 0 = 2 - 2g$. Brouwer originally defined its degree in terms of the numbers p_+, p_- of respectively positive and negative image simplexes covering the point $f(p)$. He noticed that the difference $p_+ - p_-$ is constant for regular points and defined this quantity as the degree of the continuous map f . In arbitrary dimension n , what we are doing is that we are attaching to our manifold a collection K of n -simplexes such that:

- (1) if $s \in K$, then every face of s also belongs to K ;
- (2) if $s, t \in K$, then $s \cap t$ is either empty or a common face of s and of t .

This is called a *simplicial complex*. We can give an orientation to such a construction by ordering the set of its vertices. We can also associate to each complex the Abelian group $C_n(K)$ generated by the tuples $(v_0, \dots, v_n)$ of $n + 1$ vertices with the following relation:

- (1) $(v_0, \dots, v_n) = 0$ if some vertex is repeated;
- (2) $(v_0, \dots, v_n) = (\text{sign } \sigma)(v_{\sigma(0)}, \dots, v_{\sigma(n)})$ for any permutation σ of $\{v_0, \dots, v_n\}$.

We denote by $\langle v_0, \dots, v_n \rangle$ the element of $C_n(K)$ corresponding to $(v_0, \dots, v_n)$. [Rot]

DEFINITION 1.5. We define the *boundary map* $\partial_n : C_n(K) \longrightarrow C_{n-1}(K)$ by setting

$$\partial_n(\langle v_0, \dots, v_n \rangle) = \sum_{i=0}^n (-1)^i \langle v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_n \rangle$$

and extending by linearity. One has: $\partial_{n-1} \circ \partial_n \equiv 0$ for all $n \geq 2$ and $\text{im } \partial_{n+1} \subset \ker \partial_n$.

Then $Z_n(K) = \ker \partial_n$ is the group of simplicial n -cycles, $B_n(K) = \text{im } \partial_{n+1}$ and

$$H_n(K) = Z_n(K) / B_n(K)$$

is the n^{th} (simplicial) *homology group*.

Intuitively, $H_n(K)$ measures the “holes” of dimension n in the complex: H_0 counts connected components, H_1 counts loops or tunnels, H_2 counts voids, and so on.

EXAMPLE. Consider the easiest triangulation K of a square.

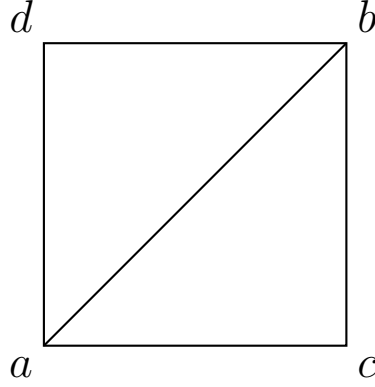


FIGURE 3. Simplicial complex associated to a square, with vertices a, b, c, d oriented alphabetically.

Then, C_2 consists of all linear combination $\mu(abc) + \lambda(abd)$ while all the higher groups are trivial. Thus, $\partial_3 \equiv 0$ i.e. $\text{im } \partial_3 = \{0\}$ while

$$\partial_2(abc) = bc - ac + ab, \quad \partial_2(abd) = bd - ad + ab$$

whence $\ker \partial_2 = \{0\}$ and thus $H_2(K) \cong 0$. Similarly, $H_1(K)$ is trivial while $H_0(K) \cong \mathbb{Z}$.

If K is an oriented simplicial complex of dimension m , then we define its *Euler-Poincaré characteristic* to be

$$\chi(K) = \sum_{n=0}^m (-1)^n \beta_n = V - E + F - \beta_3 + \beta_4 + \cdots + \beta_m$$

where β_n denotes the number of n -simplexes in K . This is equivalent to defining

$$\chi(K) = \sum_{n=0}^m (-1)^n \text{rank } H_n(K).$$

As we can always associate to a compact oriented manifold $\mathcal{M}$ an m -dimensional simplicial complex, we have a full description of the homology groups of the manifold.

POINCARÉ-HOPF THEOREM. Let $\mathcal{M}$ be a compact oriented manifold and v a smooth vector field on $\mathcal{M}$ with isolated zeros z , then

$$\sum_z \text{ind}_z(v) = \chi(\mathcal{M})$$

and thus this index sum is a topological invariant of $\mathcal{M}$, as it does not depend on the particular choice of vector field.

A specific instance of this result is the hedgehog theorem, because $\chi(\mathbb{S}^n) = 1 + (-1)^n$ implies that any vector field v on some even-dimensional sphere has at least one zero. As a another consequence, one also gets the celebrated Gauss-Bonnet formula for the integral curvature:

$$\deg(G) = \chi(\mathcal{M})$$

which, for a surface, takes the well-known form:

$$\int_{\mathcal{M}} K_p = 4\pi(1 - g).$$

More importantly, Brouwer fixed-point theorem can also be seen as a corollary to the Poincaré-Hopf formula. Because $\chi(\mathbb{D}^n) = 1$, assuming that the vector field $(x - f(x))/|x - f(x)|$ has no zeros would directly give a contradiction.

Enclosed inside the power of the Poincaré-Hopf theorem there is a whole bunch of fixed-point theorems one can derive for the most different manifolds, and one sort of has a formula for counting, or at least upper bounding, the number of fixed points.

1.2 NUMBER OF NASH EQUILIBRIA

A fundamental result in finite game theory is that in *regular games*, the number of Nash equilibria is odd. Regularity ensures that each equilibrium is non-degenerate.

THEOREM 1.6. Let G be a finite n -player game in which every Nash equilibrium is regular. Then the number of Nash equilibria of G is odd.

PROOF. Let $x_i \in \Delta_i$ denote the mixed strategy of player i , where Δ_i is the simplex of dimension $|S_i| - 1$. Let $\Delta = \prod_{i=1}^n \Delta_i$ denote the product of simplices. Consider the following *perturbed replicator dynamics* for a small parameter $\epsilon > 0$:

$$\dot{x}_{i,s_i} = x_{i,s_i} \left(u_i(s_i, x_{-i}) - u_i(x_i, x_{-i}) - n\epsilon \right) + \epsilon, \quad (\text{G})$$

where $u_i(s_i, x_{-i})$ is the expected payoff to pure strategy s_i and $u_i(x_i, x_{-i})$ is the expected payoff to the mixed strategy x_i .

Let X_ϵ denote the vector field defined by the right-hand side of (G). Along the boundary $\partial\Delta$, the vector field X_ϵ points into the simplex. Therefore, the vector field $-X_\epsilon$ points outward along the boundary. By the Poincaré–Hopf theorem, the sum of the indices of the singularities (zeros) of $-X_\epsilon$ is equal to the Euler characteristic $\chi(\Delta) = 1$. Each singularity of X_ϵ is non-degenerate for $\epsilon > 0$ sufficiently small, by regularity. Moreover, the indices satisfy

$$\text{ind}(-X_\epsilon, x_0) = (-1)^{\dim(\Delta)} \text{ind}(X_\epsilon, x_0) = (-1)^{n-1} \text{ind}(X_\epsilon, x_0),$$

since $\dim(\Delta) = n - 1$ for an n -player game. Hence, the sum of the indices of X_ϵ is $(-1)^{n-1}$. Let k be the number of singularities with index $+1$ and ℓ be the number with index -1 . Then

$$k - \ell = (-1)^{n-1}, \quad k + \ell = \text{total number of singularities}.$$

A simple parity argument shows that $k + \ell$ must be odd. Therefore, X_ϵ has an odd number of singularities.

Let $z(\epsilon)$ be any singularity of X_ϵ , i.e.

$$x_{i,s_i} \left(u_i(s_i, z_{-i}(\epsilon)) - u_i(z_i(\epsilon), z_{-i}(\epsilon)) - n\epsilon \right) + \epsilon = 0.$$

Dividing by $x_{i,s_i} > 0$ and taking the limit as $\epsilon \rightarrow 0$, we see that any limit point $\bar{z}$ satisfies

$$u_i(s_i, \bar{z}_{-i}) = u_i(\bar{z}_i, \bar{z}_{-i}), \quad \text{for all } s_i \text{ with } \bar{p}_{i,s_i} > 0,$$

which is precisely the Nash equilibrium condition. Hence, all singularities of X_0 correspond to Nash equilibria. Since X_0 is regular, its singularities are isolated. By continuity, each singularity of X_0 corresponds to a singularity of X_ϵ for small $\epsilon > 0$. Therefore, the number of Nash equilibria of G is odd. ■

2. LEFSCHETZ FIXED-POINT THEOREM

We conclude by leveraging the usefulness of homology to derive the ultimate fixed-point theorem, generalizing and implying all the previous results we have seen. [Bre]

We start by devising an actual tool for counting fixed points of a continuous map: its Lefschetz number. Let us go back to the simplicial image, and consider *simplicial maps* $\varphi : K \rightarrow K$,

those that return a simplicial structure, i.e. for any simplex $\Delta \subset K$ its image $\varphi(K)$ is still a simplex. This is equivalent to requiring that φ is affine on each simplex:

$$\varphi \left(\sum_i \lambda_i v_{\sigma_i} \right) = \sum_i \lambda_i \varphi(v_{\sigma_i})$$

so that such a map is characterized by its values on vertices.

DEFINITION 2.1. Let K be a finite simplicial complex and let $\varphi : K \rightarrow K$ be a simplicial map. For each $n \geq 0$, we define the *trace of φ in degree n* as

$$\mathrm{tr}_n(\varphi) = \sum_{\Delta^n \subset K} \mathrm{diag}_\varphi(\Delta^n),$$

where the sum runs over all oriented n -simplices Δ^n of K , and

$$\mathrm{diag}_\varphi(\Delta^n) = \begin{cases} 0, & \text{if } \varphi(\Delta^n) \cap \Delta^n \subsetneq \Delta^n, \\ +1, & \text{if } \varphi(\Delta^n) = \Delta^n \text{ and } \varphi \text{ preserves the orientation of } \Delta^n, \\ -1, & \text{if } \varphi(\Delta^n) = \Delta^n \text{ and } \varphi \text{ reverses the orientation of } \Delta^n. \end{cases}$$

The *Lefschetz number* of φ is then defined as

$$\Lambda_\varphi = \sum_{n \geq 0} (-1)^n \mathrm{tr}_n(\varphi).$$

EXAMPLE. Consider the 1-dimensional simplicial complex $K = \partial\Delta$ given by one triangle with vertices a, b, c . It is easy to check that $H_0(K), H_1(K) \cong \mathbb{Z}$. Consider the simplicial map $\varphi : K \rightarrow K$ sending each vertex into the successive one, then $\mathrm{tr}_0(\varphi) = \mathrm{tr}_1(\varphi) = 0$ and thus $\Lambda_\varphi = 0$, as this map effectively has no fixed points. This map induces the homomorphisms $\varphi_*^{(n)} : C_n(K) \rightarrow C_n(K)$, where for instance

$$\varphi_*^{(1)}(a) = b, \quad \varphi_*^{(1)}(b) = c, \quad \varphi_*^{(1)}(c) = a$$

so it can be represented by the matrix

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{with trace} = 0,$$

whence $\text{tr}_1(\varphi) = \text{tr } \varphi_*^{(1)} = 0$. Notice that we can associate the simplicial complex $K = \partial\Delta$ considered here to the circle $\mathbb{S}^1$, and we recover theorem 3.2. In fact, given any simplicial map $\varphi : \partial\Delta \longrightarrow \partial\Delta$, because our space is connected and thus has only one connected component, $\varphi_*^{(0)} \equiv \text{Id}_1$ so $\text{tr } \varphi_*^{(0)} = 1$ while $C_1(\mathbb{S}^1)$ is the group generated by the standard loop that goes around the circle once counter-clockwise and any homomorphism on it is fully determined by the winding number w of times it wraps around the circle, as it takes the generator and scales it for this number, whence $\text{tr } \varphi_*^{(1)} = w = \deg(\varphi)$ and thus

$$\Lambda_\varphi = \text{tr } \varphi_*^{(0)} - \text{tr } \varphi_*^{(1)} = 1 - \deg(\varphi)$$

because all the higher homology groups are trivial and the traces vanish, so indeed if

$$\Lambda_\varphi = 1 - \deg(\varphi) \neq 0$$

then the map φ has fixed points. We will make precise how to infer the same conclusion for the corresponding continuous map $f : \mathbb{S}^1 \longrightarrow \mathbb{S}^1$.

Given a simplicial map $\varphi : K \longrightarrow K$ and the homomorphisms $\varphi_*^{(n)} : C_n(K) \longrightarrow C_n(K)$ it induces, we can indeed equivalently define

$$\Lambda_\varphi = \sum_{n \geq 0} (-1)^n \text{tr } \varphi_*^{(n)}.$$

Up to refining our simplicial complex structure, we can always approximate any continuous map $f : K \longrightarrow K$ by means of some simplicial map $\ddagger \tilde{f} : K \longrightarrow K$.

More precisely, for any $\epsilon > 0$, we can always find a simplicial map that is homotopically equivalent to our continuous map, i.e. $f \sim \tilde{f}$ and $d(f(p), \tilde{f}(p)) < \epsilon$ for all $p \in K$.

[‡]Using the terms introduced here, the lemma of Sperner from Chapter I can be equivalently stated as follows: if K is any subdivision of Δ^n and $\varphi : K \longrightarrow \Delta^n$ is a simplicial approximation to the identity, then the number of simplices of K that map onto Δ^n is odd.

We can thus define the *Lefschetz number* of the continuous map as

$$\Lambda_f = \Lambda_{\tilde{f}} = \sum_{n \geq 0} (-1)^n \operatorname{tr} \tilde{f}_*^{(n)}$$

as it goes without saying that the Lefschetz number is a homotopic invariant. However, we also have that

$$\Lambda_f = \sum_{n \geq 0} (-1)^n \operatorname{tr} f_*^{(n)}$$

for the induced homomorphisms $f_* : H_n(K) \longrightarrow H_n(K)$ so, in particular

$$\Lambda_{\operatorname{Id}} = \sum_{n \geq 0} (-1)^n \operatorname{rank} H_n(K) = \chi(K).$$

LEFSCHETZ FIXED-POINT THEOREM. Let K be a compact simplicial complex and $f : K \longrightarrow K$ a continuous map with Lefschetz number $\Lambda_f \neq 0$. Then, f has a fixed point.

PROOF. Consider any simplicial approximation $\tilde{f} \sim f$ and assume $\tilde{f}$ (and equivalently f) has no fixed points, then we want to show that $0 = \Lambda_{\tilde{f}} = \Lambda_f$.

Notice that for a simplicial map, only simplices that are mapped onto themselves contribute to the trace $\operatorname{tr}_n(\tilde{f}) = \sum_{\Delta^n \subset K} \operatorname{diag}_{\tilde{f}}(\Delta^n)$, but whenever $\tilde{f}(\Delta) = \Delta$, because $\tilde{f}$ is affine on each simplex, it has a fixed point but this is a contradiction so we must always have

$$\tilde{f}(\Delta) \neq \Delta \quad \text{for every simplex } \Delta \subset K$$

so in particular $\operatorname{diag}_{\tilde{f}} \equiv 0 = \Lambda_{\tilde{f}}$. ■

Let us now consider the case of a smooth map $f : \mathcal{M} \longrightarrow \mathcal{M}$ and denote by $\operatorname{Fix}(f)$ the nonempty set of its non-degenerate fixed points. Then,

$$\Lambda_f = \sum_{p \in \operatorname{Fix}(f)} \operatorname{ind}(f) = \sum_{p \in \operatorname{Fix}(f)} \deg(\operatorname{Id} - f) = \sum_{p \in \operatorname{Fix}(f)} \operatorname{sign} \det(\operatorname{Id} - df_p).$$

This is the formula counting fixed points we have so long craved for.

APPENDIX A

APPLICATIONS OF THE BROUWER FIXED-POINT THEOREM

1. THE GAME OF HEX

We can model the Hex game board via the graph B_k , with $k \in \mathbb{N}$, whose vertices are defined as follows:

$$V(B_k) = \{z \in \mathbb{Z}^2 : 1 \leq z_i \leq k, i = 1, 2\}$$

and are connected by an edge iff $|z - \tilde{z}| = 1$ and they are comparable, that is:

$$z_i \leq \tilde{z}_i \quad \forall i = 1, 2 \quad \text{or viceversa.}$$

Assume Player 1 (horizontal) wins if they are able to connect the boundaries $W = (\{0\} \times [0, k]) \cap B_k$ and $E = (\{k\} \times [0, k]) \cap B_k$, while Player 2 (vertical) wins if they are able to connect boundaries $S = ([0, k] \times \{0\}) \cap B_k$ and $N = ([0, k] \times \{k\}) \cap B_k$. Then, to prove that Hex can never end in a draw, we would like to prove the following result.

THEOREM 1.1. Let B_k be (vertex)-covered by two sets H, V , then either H contains a connected set meeting E and W or V contains a connected set meeting N and S .

Instead, we will prove that this is implied by the following version of Brouwer's fixed point theorem. [Gal79]

THEOREM 1.2 (BROUWER). Let $I_k^2 = [0, k]^2$ denote a square.

Given any continuous mapping $f : I_k^2 \longrightarrow I_k^2$, there exists a fixed point $x \in I^2 : f(x) = x$.

In fact $\mathbb{D}^2$ and I^2 are homeomorphic.

LEMMA 1.3. Any function $f : B_k \longrightarrow \mathbb{R}^2$ extends to a *simplicial mapping* $\tilde{f} : I_k^2 \longrightarrow \mathbb{R}^2$, that is, $\tilde{f}$ is continuous and for all $\{\lambda_i\}_i$ s.t. $\sum_i \lambda_i = 1$:

$$\tilde{f} \left(\sum_i \lambda_i x_i \right) = \sum_i \lambda_i \tilde{f}(x_i)$$

PROOF. Given any 2-simplex in I_k^2 , any point inside it can be written as a convex combination of its vertices:

$$x = \sum_i \lambda_i x_i, \quad \text{where } \sum_i \lambda_i = 1, \quad \lambda_i \geq 0$$

and to extend f to all of I_k^2 , we define $\tilde{f}$ at any such x by the formula:

$$\tilde{f}(x) = \sum_i \lambda_i f(x_i)$$

which uniquely determines $\tilde{f}$ in a piecewise affine manner over each simplex:

$$\tilde{f} \left(\sum_i \lambda_i x_i \right) = \sum_i \lambda_i \tilde{f}(x_i)$$

and, since it is affine, $\tilde{f}$ remains continuous within each simplex. ■

LEMMA 1.4. Let z_1, z_2, z_3 be the vertices of any 2-simplex and

$$\rho(z_i) := z_i + v_i$$

for given vectors v_1, v_2, v_3 . Then, the simplicial extension $\tilde{\rho}$ has a fixed point if and only if $0 \in \text{co}\{v_1, v_2, v_3\}$.

PROOF. Let $x = \lambda_1 z_1 + \lambda_2 z_2 + \lambda_3 z_3$ then

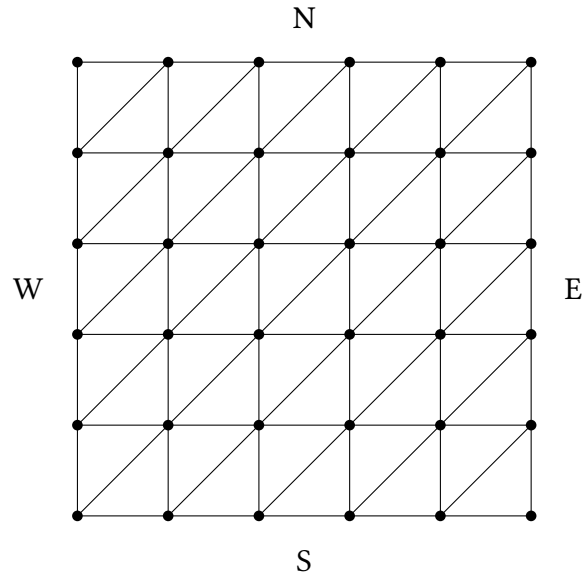
$$\begin{aligned} \tilde{\rho}(x) &= \lambda_1(z_1 + v_1) + \lambda_2(z_2 + v_2) + \lambda_3(z_3 + v_3) \\ &= x + \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 \end{aligned}$$

We can see that for a fixed point to exist there must be suitable $\lambda_1, \lambda_2, \lambda_3$ such that the convex combination of the v_1, v_2, v_3 with these coefficients gives 0.

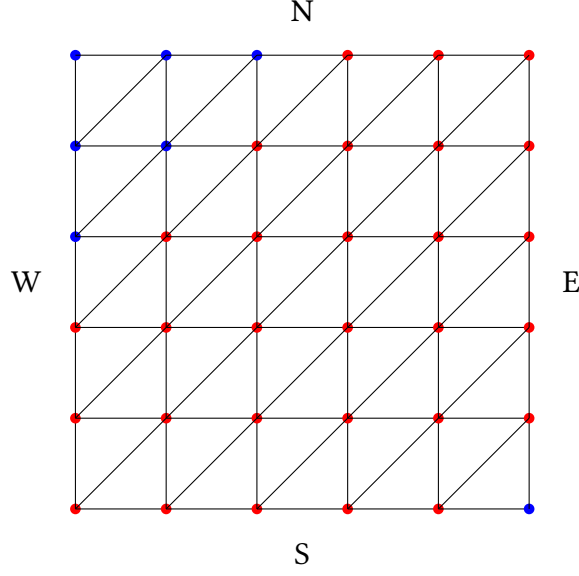
That is, $0 \in \text{co}\{v_1, v_2, v_3\}$. ■

THEOREM 1.5. Brouwer implies Hex.

PROOF. By Carathéodory's theorem, each point of I^2 is in the convex hull of at most three vertices, all of which are connected by an edge. These being, the triangles in B_k , i.e., edges and vertices in the grid:



Now, assume B_k is partitioned into two sets $H \cup V = B_k$, and let $\tilde{W}$ be the set of vertices connected to W via a path in H and $\tilde{E} = H \setminus \tilde{W}$, and similarly, $\tilde{S}$ be the set of vertices connected to S via a path in V and $\tilde{N} = V \setminus \tilde{S}$. Clearly, there is no edge connecting $\tilde{W}$ and $\tilde{E}$, as well as there is no edge connecting $\tilde{S}$ and $\tilde{N}$. As an example of this fact, consider the following partition:



where H consists of blue vertices and V of the red ones. Then, in this case, $\tilde{W}$ is composed of the vertices in the leftmost upper angle, while $\tilde{E}$ is the single vertex in the rightmost lower angle. Clearly, these two components of H are disconnected.

By contradiction, suppose there exists no path in H that connects E to W and no path in V from N to S .

Consider the function $f : B_k \longrightarrow B_k$ defined as follows:

$$f(z) := \begin{cases} z + e_1 & \text{if } z \in \tilde{W} \\ z - e_1 & \text{if } z \in \tilde{E} \\ z + e_2 & \text{if } z \in \tilde{S} \\ z - e_2 & \text{if } z \in \tilde{N} \end{cases}$$

which is well-defined because:

- (1) $z + e_1 \notin B_k$ only if $z \in E$, but by assumption, there is no path in H from W to E , hence $E \cap \tilde{W} = \emptyset$ i.e. $z \notin \tilde{W}$.
- (2) $z - e_1 \notin B_k$ only if $z \in W$, but by assumption, there is no path in H from E to W , hence $W \cap \tilde{E} = \emptyset$ i.e. $z \notin \tilde{E}$.

- (3) $z + e_2 \notin B_k$ only if $z \in N$, but by assumption, there is no path in V from S to N , hence $N \cap \tilde{S} = \emptyset$ i.e. $z \notin \tilde{S}$.
- (4) $z - e_2 \notin B_k$ only if $z \in S$, but by assumption, there is no path in V from N to S , hence $S \cap \tilde{N} = \emptyset$ i.e. $z \notin \tilde{N}$.

By the lemmas, we can extend f simplicially to I_k^2 .

W.t.s. f has no fixed point (contradicting Brouwer).

Because $\tilde{W}$ and $\tilde{E}$ are connected and neither $\tilde{N}$ and $\tilde{S}$ are, for any triangle, as before, it cannot happen that a vertex gets shifted right/up $(+e_1, +e_2)$ and one other vertex gets shifted left/down $(-e_1, -e_2)$, in the same triangle, meaning that the three vertices are translated by vectors which lie in a single quadrant of $\mathbb{R}^2$, i.e. $0 \notin \text{co}(\text{vertices})$, contradicting the condition for the existence of a fixed point in the previous lemma. ■

2. JORDAN CURVE THEOREM

The following was proved in [Mae84].

PATH-INTERSECTION LEMMA. Let $f, g : I \longrightarrow I \times I$ be continuous maps, where $I = [0, 1]$. Assume

$$f(0) = (a, 0), \quad f(1) = (b, 1), \quad g(0) = (0, c), \quad g(1) = (1, d)$$

for some $a, b, c, d \in I$. Then there exist $s, t \in I$ such that $f(s) = g(t)$.

PROOF. Define a continuous map $h : I \times I \longrightarrow I \times I$ as follows. Let

$$f(t) = (f_1(t), f_2(t)), \quad g(t) = (g_1(t), g_2(t)).$$

Set

$$h(x, y) = (g_1^{-1}(f_2(x)), f_2^{-1}(g_1(y))),$$

where g_1^{-1} denotes a continuous inverse along g_1 (using monotone reparametrization if necessary). Intuitively, $h(x, y)$ “slides” the coordinates along the paths so that $f(s)$ and $g(t)$ align. Observe that h maps $I \times I$ into itself: $h(I \times I) \subset I \times I$, and is continuous. By the Brouwer fixed-point theorem, there exists a fixed point $(s, t) \in I \times I$ such that $h(s, t) = (s, t)$.

By construction of h , this fixed point satisfies $f(s) = g(t)$, which proves the lemma. ■

JORDAN CURVE THEOREM. Let $\Gamma \subset \mathbb{R}^2$ be a simple closed curve. Then $\mathbb{R}^2 \setminus \Gamma$ has exactly two connected components: an interior and an exterior, and Γ is the boundary of both.

SKETCH OF THE PROOF. Let Γ be parametrized by a continuous map $\gamma : S^1 \longrightarrow \mathbb{R}^2$. To show that Γ separates the plane, take any continuous path connecting a point inside the curve to a point outside. By compactness and continuity, we can approximate this path using finitely many straight segments forming paths across a square containing the curve. We apply Maehara's path-intersection lemma to each pair of segments crossing the square: any path from left to right must intersect any path from bottom to top within the square. Iterating this argument for covering squares shows that any path from inside to outside must intersect γ at some point.

Thus, Γ separates the plane into exactly two connected regions. ■

3. PERRON–FROBENIUS THEOREM

PERRON–FROBENIUS THEOREM. Let $A = (a_{ij})_{i,j=1}^n$ be an $n \times n$ matrix with $a_{ij} > 0$ for all $i, j = 1, \dots, n$. Then there exists a positive eigenvalue $\lambda > 0$ and a corresponding eigenvector $v \in \mathbb{R}^n$ with $v_i > 0$ for all $i = 1, \dots, n$.

PROOF. Let $\Delta^{n-1} = \{v \in \mathbb{R}^n : v_i \geq 0, \sum_{i=1}^n v_i = 1\}$, which is a compact, convex subset of $\mathbb{R}^n$. Define

$$\sigma(v) = \sum_{i=1}^n v_i, \quad g(v) = \frac{Av}{\sigma(Av)} \quad \text{for } v \in \Delta^{n-1}.$$

so that $g(v) \in \Delta^{n-1}$ since all entries of Ax are positive and dividing by the sum normalizes it and thus $g : \Delta^{n-1} \longrightarrow \Delta^{n-1}$ is continuous.

Since Δ^{n-1} is compact and convex and g is continuous, Brouwer's fixed-point theorem guarantees that there exists $v^* \in \Delta^{n-1}$ such that $g(v^*) = v^*$.

By definition of g , the fixed point satisfies

$$v^* = g(v^*) = \frac{Av^*}{\sigma(Av^*)} \implies Av^* = \sigma(Av^*) v^*.$$

Set $\lambda = \sigma(Av^*) > 0$. Then $v^* \in \Delta^{n-1}$ is a positive eigenvector and λ is a positive eigenvalue and, since all entries of A are strictly positive, $Av^* > 0$ entrywise. Therefore, $v^* > 0$ entrywise as well. ■

This result can itself be used in applications for some instances of the fixed-point problem.

EXAMPLE. Consider a Markov chain $\{X_t\}_{t \geq 0} \subset S$ with transition probabilities

$$P_{ij} = \mathbb{P}\{X_{t+1} = j | X_t = i\}$$

stored in some matrix $P = (P_{ij})_{i,j \in S}$, then finding a stationary distribution for $\{X_t\}_t$ is equivalent to solving the fixed-point problem:

$$\pi = \pi P$$

for some distribution probability distribution π , that is:

$$\pi_j = \sum_{i \in S} \pi_i P_{ij}$$

for all $i, j \in S$. The interpretation is that, because $\sum_i \pi_i P_{ij}$ is the probability of $X_{t+1} = j$ given that $X_t \sim \pi$, then π is a stationary distribution for $\{X_t\}_t$ if, given a starting state $X_0 \sim \pi$ it follows $X_1 \sim \pi$, and by induction $X_t \sim \pi$ for all $t \geq 0$: requiring π to be a stationary distribution for a Markov chain $\{X_t\}_t$ is equivalent to require that

$$X_0 \sim \pi \implies X_t \sim \pi, \quad \text{for all } t \geq 0.$$

When the state space S of the stochastic process is finite: $|S| = n < +\infty$, then a probability distribution $\pi \in \Delta^{n-1}$ is stationary if and only if it is a left eigenvector of P with eigenvalue 1: the existence is guaranteed by the Perron–Frobenius theorem.

APPENDIX B

PROOF OF SARD THEOREM

Recall that we refer to a set $S \subset \mathbb{R}^n$ to have a *measure zero* if it can be covered by countably many cubes, or balls, of arbitrarily small volume: more precisely, for all $\epsilon > 0$, there exists an at most countable collection $\{C_i\}_i$ s.t. $S \subset \bigcup_i C_i$ and $\sum_i \text{vol}(C_i) > \epsilon$.

In the following, we let $f : \mathcal{M} \longrightarrow \mathbb{R}^n$ be a smooth function. First, we notice that being of measure zero is a smooth invariant.

LEMMA. Let $\mathcal{M} \subset \mathbb{R}^m$ be just an open set and $S \subset \mathcal{M}$ be of measure zero, then $f(S) \subset \mathbb{R}^n$ has measure zero.

Assuming moreover compactness of the set S , one also has the following sufficient condition for it being of measure zero.

FUBINI'S THEOREM. Given a compact $S \subset \mathbb{R}^n = \mathbb{R} \times \mathbb{R}^{n-1}$, if its intersection $S \cap \mathcal{P}_t$, with each hyperplane $\mathcal{P}_t = \{t\} \times \mathbb{R}^{n-1}$, has measure zero, then S has measure zero.

More precisely, our definition of being a set of measure zero is equivalent to writing $\mathcal{L}^n\{S\} = 0$, where $\mathcal{L}^n$ denotes the n -dimensional Lebesgue measure. Then, Fubini's theorem identifies in the following:

$$\mathcal{L}^{n-1}\{S \cap \mathcal{P}_t\} = 0 \quad \text{for all } t \in \mathbb{R},$$

a sufficient condition for the n -dimensional negligibility of the set S itself.

Let $\mathcal{D}_f \subset \mathcal{C}_f$ be the set of points $p \in \mathcal{M}$ at which df_p vanishes.

LEMMA. Given any open set $\Omega \subset \mathbb{R}^m$ and any smooth $\varphi : \Omega \longrightarrow \mathbb{R}$, then $\varphi(\mathcal{D}_\varphi)$ has measure zero in $\mathbb{R}$.

PROOF. Define a filtration of $\mathcal{D}_\varphi$ given by $\mathcal{D}_k = \{p \in \Omega : d^\alpha f_p = 0 \text{ for all } |\alpha| \leq k\}$,

$$\mathcal{D}_\varphi \equiv \mathcal{D}_1 \supset \mathcal{D}_2 \supset \mathcal{D}_3 \supset \cdots \supset \mathcal{D}_m.$$

We argue by induction on k , from the base case $k = m$ downwards.

BASE CASE. All partial derivatives of φ up to order m vanish at $p \in \mathcal{D}_m$, that is,

$$d^\alpha \varphi_p = 0 \quad \text{for all } \alpha = (\alpha_1, \dots, \alpha_k) \text{ s.t. } |\alpha| \leq k.$$

whence by the expansion:

$$\varphi(x) = \varphi(p) + \sum_{|\alpha|=m+1} \frac{1}{\alpha!} (x-p)^\alpha d^\alpha \varphi_\xi,$$

we obtain $|\varphi(x) - \varphi(p)| \lesssim \|x - p\|^{m+1}$. Let $Q \subset \Omega$ be a cube of side length δ containing p . Then, for all $x, y \in Q \cap \mathcal{D}_m$,

$$|\varphi(x) - \varphi(y)| \leq C \delta^{m+1}.$$

Hence the image $\varphi(Q \cap \mathcal{D}_m)$ is contained in an interval of length at most $C\delta^{m+1}$.

Now, cover $\mathcal{D}_m$ by a countable collection of such cubes $\{Q_i\}_i$ with side lengths δ_i . Then $\varphi(\mathcal{D}_m) \subset \bigcup_i \varphi(Q_i \cap \mathcal{D}_m)$, and therefore

$$\mathcal{L}^1\{\varphi(\mathcal{D}_m)\} \leq \sum_i C \delta_i^{m+1}.$$

Since we can choose the covering such that $\sum_i \delta_i^m < \infty$, it follows that the right-hand side can be made arbitrarily small, $\mathcal{L}^1\{\varphi(\mathcal{D}_m)\} = 0$.

INDUCTIVE STEP. Assume that for some $k \leq m$, we have that $\varphi(\mathcal{D}_k)$ is negligible, then we want to show that $\varphi(\mathcal{D}_{k-1})$ also has measure zero. By definition, for each point $p \in \mathcal{D}_{k-1} \setminus \mathcal{D}_k$, there exists a multi-index α with $|\alpha| = k$ such that $d^\alpha \varphi(p) \neq 0$.

We want to show $\varphi(\mathcal{D}_{k-1} \setminus \mathcal{D}_k)$ has measure zero.

Assume that $\mathcal{D}_k = \emptyset$ (or has been removed from a neighborhood $\mathcal{U}$), for each point $p \in \mathcal{D}_{k-1} = \mathcal{D}_{k-1} \setminus \mathcal{D}_k$, there exists a multi-index α with $|\alpha| = k$ such that $d^\alpha \varphi(p) \neq 0$, namely that all derivatives of order less than k vanish, but there exists at least one derivative of order k which does not vanish. Let ψ be such an $k - 1^{\text{th}}$ order derivative of φ , so that its differential $d\psi_p \neq 0$. Then $\psi = 0$ on $\mathcal{D}_{k-1}$, and that is a regular value of ψ . Define

$$\mathcal{V} = \psi^{-1}(0) \cap \mathcal{U}.$$

By the Implicit Function Theorem, $\mathcal{V}$ is a smooth submanifold of codimension one in U . Moreover, $\mathcal{D}_{k-1} \subset V$ because all points in $\mathcal{D}_{k-1}$ satisfy $\psi = 0$, and $\mathcal{D}_{k-1}$ is contained in the set of critical points of $\varphi|_{\mathcal{V}}$ since the differential of φ vanishes on $\mathcal{D}_{k-1}$.

By the inductive hypothesis on dimension, the image

$$\varphi(\mathcal{D}_{k-1}) = \varphi|_{\mathcal{V}}(\mathcal{D}_{k-1})$$

has measure zero. ■

LEMMA. $f(\mathcal{D}_f)$ has measure zero in $\mathbb{R}^n$.

PROOF. Because we are considering $f : \mathcal{M} \rightarrow \mathbb{R}^n$, we can write $f = (f^{(1)}, \dots, f^{(n)})$ where each $f^{(\ell)} : \mathcal{M} \rightarrow \mathbb{R}$ is a smooth function, $\ell = 1, \dots, n$.

Let $p \in \mathcal{D}_f$, i.e. $df_p = 0$, then each $df_p^{(\ell)} = 0$ and thus $f(\mathcal{D}_f) \subset f^{(\ell)}(\mathcal{D}_{f^{(\ell)}}) \times \mathbb{R}^{n-1}$. Because $\mathcal{L}^n\{f(\mathcal{D}_f)\} = \mathcal{L}^1\{f^{(\ell)}(\mathcal{D}_{f^{(\ell)}})\}\mathcal{L}^{n-1}\{\mathbb{R}^{n-1}\}$, then it suffices to show our claim for a real-valued function, that is for the case $n = 1$, which is covered by the previous lemma. ■

LEMMA. $f(\mathcal{C}_f \setminus \mathcal{D}_f)$ has measure zero in $\mathbb{R}^n$.

PROOF. Since $\mathcal{D}_f$ is closed, we may remove it from the domain without loss of generality and assume that $\mathcal{D} = \emptyset$. Thus, at every point $p \in \mathcal{C}_f$, the differential df_p is nonzero.

Hence, there exists a coordinate projection $\pi : \mathbb{R}^k \rightarrow \mathbb{R}$ (say, projection onto the last coordinate) such that the differential of $\pi \circ f$ does not vanish at p . By continuity, there is an open neighborhood $\mathcal{V}$ of p in which $d\pi \circ df$ is nowhere zero. Therefore, all values are regular for the map

$$\pi \circ f : \mathcal{V} \rightarrow \mathbb{R}.$$

For each $t \in \mathbb{R}$, define the level set

$$\mathcal{V}_t = (\pi \circ f)^{-1}(t).$$

Since t is a regular value, each $\mathcal{V}_t$ is a smooth $(n-1)$ -dimensional submanifold of $\mathcal{V}$.

Now define the restricted map

$$f_t = f|_{\mathcal{V}_t} : \mathcal{V}_t \rightarrow \mathbb{R}^{k-1} \times \{t\}.$$

Observe that at any point $p \in \mathcal{V}_t$, the differential df_p maps some vector to a nonzero component in the $\mathbb{R}$ -direction (the direction corresponding to π). Consequently, p is critical for f if and only if it is for f_t . Thus,

$$\mathcal{C}_f \cap \mathcal{V}_t = \mathcal{C}_{f_t}.$$

By the inductive hypothesis on the dimension of the domain, for each fixed t , the image

$$f_t(\mathcal{C}_{f_t}) = f(\mathcal{C}_f \cap \mathcal{V}_t)$$

has measure zero in the hyperplane $\mathbb{R}^{k-1} \times \{t\}$.

Now, to show that $f(\mathcal{C}_f \cap \mathcal{V})$ has measure zero in $\mathbb{R}^k$, note that it suffices to verify that the image of $\mathcal{C}_f \cap Q$ has measure zero for each cube $Q \subset V$, by Fubini's theorem. ■

THEOREM. If $f : \mathcal{M} \longrightarrow \mathbb{R}^n$ is a smooth map, then its set of critical values $f(\mathcal{C}_f)$ has measure zero.

It is not difficult to derive the general statement from the previous one.

We mainly adapted discussions from [Mil; Bre].

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